

Customer Shopping Behavior Analysis

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Data Analysis Project | Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Key features: demographics, purchase details, shopping behavior



Data Preparation in Python

→ Load & Explore

- Imported **3,900 transaction records** using pandas
- Used `df.info()` to identify data types and missing values
- Applied `df.describe()` for statistical summary (mean, median, std dev, quartiles)
- Identified **37 missing values** in Review Rating column only

→ Clean Missing Data

- Applied **median imputation** strategy grouped by product category
- Preserved category-specific rating patterns during imputation
- Validated imputation quality with before/after distribution analysis

→ Standardize

- Converted all column names to `snake_case` for consistency
- Removed redundant `promo_code_used` column (duplicate information)
- Ensured data type consistency across all features
- Final dataset: **3,900 rows × 18 columns**, ready for analysis

Feature Engineering

Age Group

Binned customer ages into **4 segments** using `pd.cut()` for equal-width binning:

- **Young Adult** (18–35)
- **Middle-aged** (36–50)
- **Adult** (51–65)
- **Senior** (65+)

Purchase Frequency

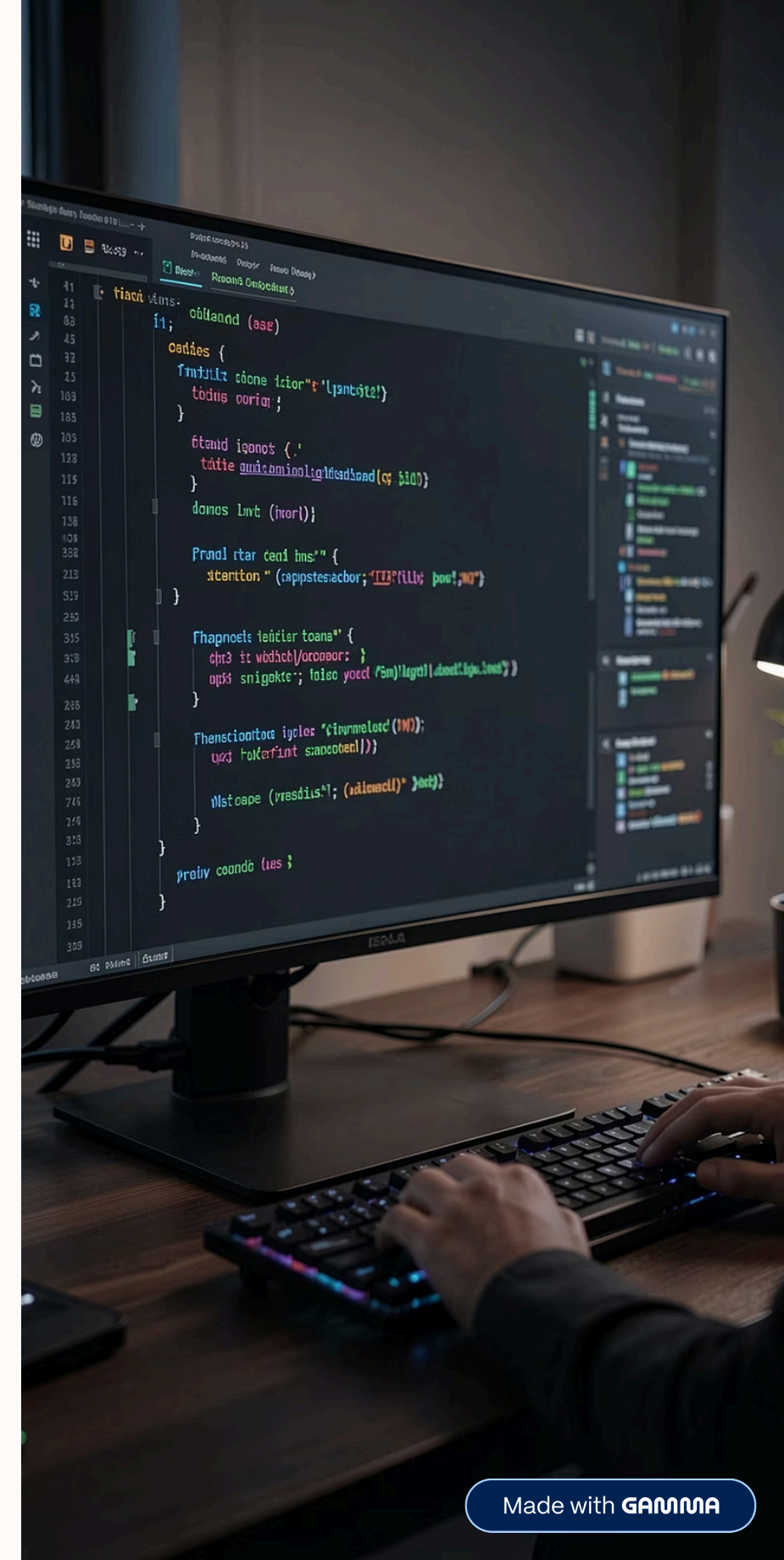
Calculated days between consecutive purchases per customer to derive behavioral metrics:

- Average purchase interval & frequency score
- Identified **repeat buyers** (5+ purchases) vs. **one-time buyers**

PostgreSQL Load

Exported cleaned DataFrame to a production-ready PostgreSQL database:

- Created **indexed tables** for optimized query performance
- Enabled real-time SQL analysis & dashboard connectivity



SQL Analysis: Revenue & Discounts

Revenue by Gender

Compared total revenue: male vs. female customers

Female: \$75,191 | **Male:** \$157,890

Subscribers vs. Non-Subscribers

Comparison of customer count, average spend, and total revenue by subscription status

Subscribers (Yes): 1,053 customers
| Avg. Spend: \$59.49 | Total Revenue: \$62,645.00

Non-Subscribers (No): 2,847 customers
| Avg. Spend: \$59.87 | Total Revenue: \$170,436.00

Discount-Dependent Products

Top 5 products with highest % of discounted purchases

1. **Hat** — 50.00%
2. **Sneakers** — 49.66%
3. **Coat** — 49.07%
4. **Sweater** — 48.17%
5. **Pants** — 47.37%

SQL Analysis: Products & Shipping

Top 5 Products by Rating

Highest average review ratings:

1. Gloves — **3.86**
2. Sandals — **3.84**
3. Boots — **3.82**
4. Hat — **3.80**
5. Skirt — **3.78**

Shipping Comparison

Average spend by shipping type:

Standard: \$58.46 | **Express:** \$60.48

Top 3 per Category

Most purchased products by category:

Accessories:

1. Jewelry — **171** orders
2. Sunglasses — **161** orders
3. Belt — **161** orders

Clothing:

1. Blouse — **171** orders
2. Pants — **171** orders
3. Shirt — **169** orders

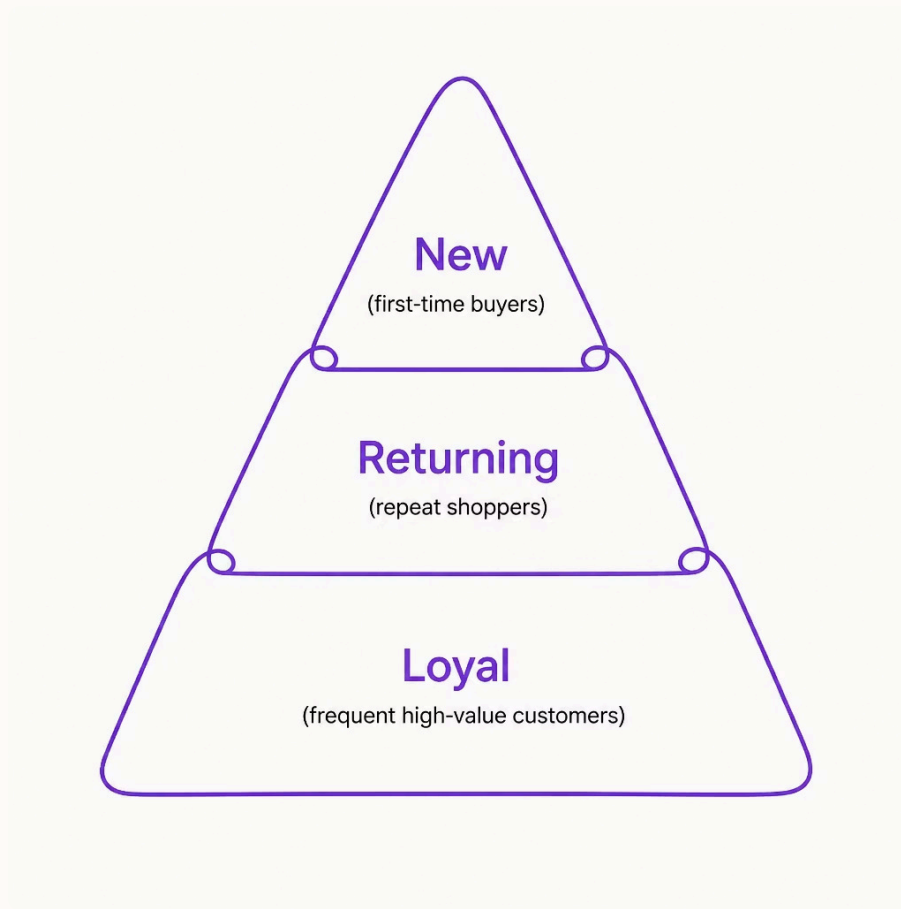
Footwear:

1. Sandals — **160** orders
2. Shoes — **150** orders
3. Sneakers — **145** orders

Outerwear:

1. Jacket — **163** orders
2. Coat — **161** orders

Customer Segmentation



Segments Defined by Purchase History

- **New** — first-time buyers
- **Returning** — repeat shoppers
- **Loyal** — frequent, high-value customers

Also analyzed: subscribers vs. non-subscribers spend & revenue

Age Groups & Subscription Behavior

Revenue by Age Group

- **Young Adult:** \$62,143
- **Middle-aged:** \$59,197
- **Adult:** \$55,978
- **Senior:** \$55,763

Young Adults generate the highest total revenue

Repeat Buyers → Subscriptions

Of customers with 5+ purchases:

- **Non-Subscribers:** 2,518 repeat buyers (73%)
- **Subscribers:** 958 repeat buyers (27%)

Majority of repeat buyers remain non-subscribers

Power BI Dashboard

Interactive visual summary of all insights — revenue, segments, products, and behavior patterns in one view.



Dashboard Features & Capabilities

This comprehensive Power BI dashboard consolidates all analytical insights into a single, dynamic view, enabling real-time exploration of revenue, customer behavior, and product performance.

1

Revenue & KPI Cards

- Total revenue, transaction count, and customer segment totals at a glance
- Real-time tracking by gender, age group, and subscription status

2

Product Performance

- Ratings, order volumes, and discount rates per product
- Category-level breakdown of top and underperforming items

3

Customer Segmentation

- Visual breakdown of New, Returning, and Loyal segments
- Shipping method analysis and its impact on average spend

4

Interactive Filters

- Drill down by category, age group, and customer type
- Slicers enable cross-filtering across all visuals simultaneously

Key Insights from Dashboard Analysis

Three critical findings that drive business strategy:

Subscription Opportunity

73% of customers shop without subscription — representing a massive untapped revenue stream. Converting even a portion of these 2,847 non-subscribers could significantly boost recurring revenue and customer lifetime value.

Clothing Category Dominance

The Clothing category leads in both revenue and sales volume, with Blouse and Pants each generating 171 orders. This category should be prioritized for inventory, marketing, and product development investments.

Young Adult Revenue Leadership

Young Adults generate the highest revenue at \$62,143 — outpacing Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763) segments. This demographic should be the primary target for marketing campaigns and product positioning strategies.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits. Only **27%** of customers are subscribers — target the **2,518 non-subscriber repeat buyers** with dedicated conversion campaigns.



Loyalty Programs

Reward repeat buyers in top segments. **Young Adults (\$62K)** and **Middle-aged (\$59K)** lead in revenue — reward frequent purchasers to increase lifetime value.



Review Discounts

Balance sales lift with margin control. **Hats (50%)**, **Sneakers (49.7%)**, and **Coats (49%)** show high discount dependency — closely monitor margin impact.



Targeted Marketing

Focus on high-value segments. **Young Adults generate the highest revenue**; Express shipping users spend **\$2 more per order** on average — prioritize these audiences.



Product Positioning

Highlight top-rated & best-selling items. **Gloves (3.86★)**, **Sandals (3.84★)**, and **Boots (3.82★)** top ratings; **Jewelry & Blouse** lead in order volume.